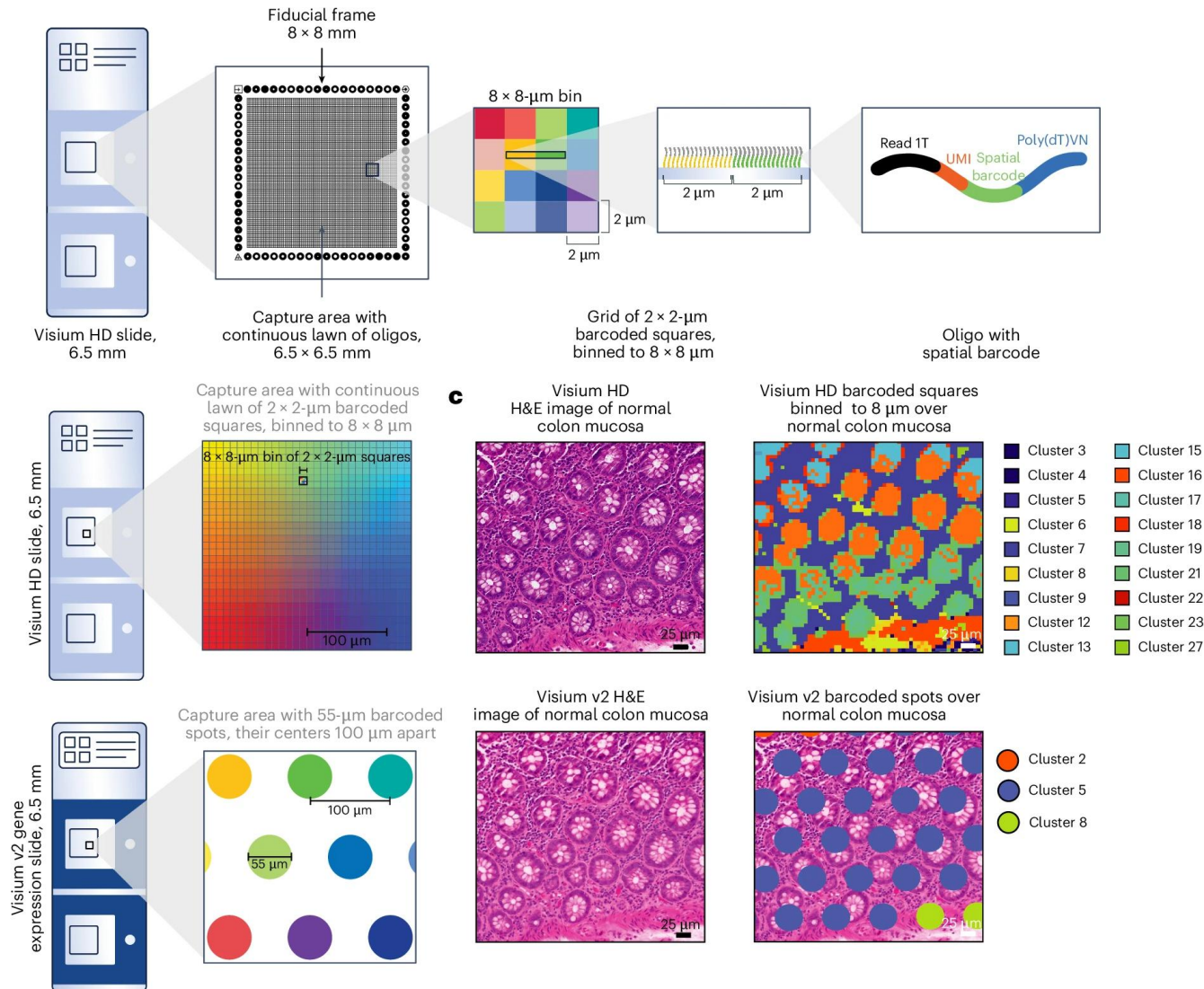


# Sequence-based spatial transcriptomics data analysis

Practical 4

Leslie Solorzano

# Visium HD



Visium HD uses a glass slide coated with a dense array of barcoded capture probes (oligonucleotides) arranged in a grid of 2×2 μm bins. Each bin contains unique spatial barcodes that label RNA molecules originating from that exact location.

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## High-definition spatial transcriptomic profiling of immune cell populations in colorectal cancer

[Michelli Faria de Oliveira](#), [Juan Pablo Romero](#), [Meii Chung](#), [Stephen R. Williams](#), [Andrew D. Gottscho](#), [Anushka Gupta](#), [Susan E. Pilipauskas](#), [Seayar Mohabbat](#), [Nandhini Raman](#), [David J. Sukovich](#), [David M. Patterson](#), [Visium HD Development Team](#) & [Sarah E. B. Taylor](#) ✉

[Nature Genetics](#) 57, 1512–1523 (2025) | [Cite this article](#)

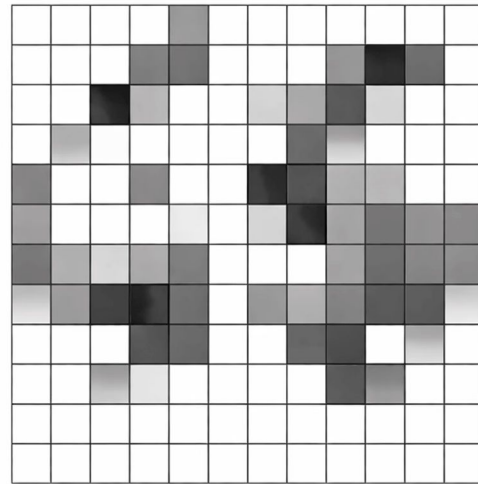


# On the grid

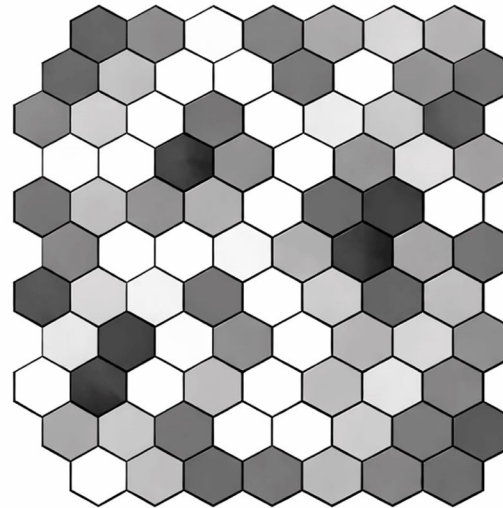
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Spatial means that spots, cells, items, anything we study, is in a grid, with coordinates, we can see where something is and what it is surrounded by. The context will allow us to study and characterize whatever we want, cells, tissue types, niches, etc.

Square grid

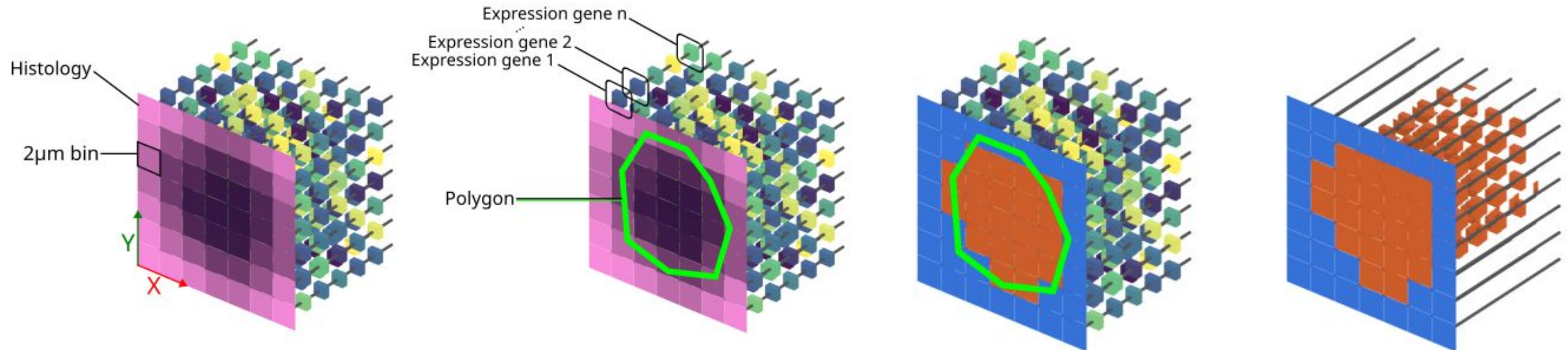


Hexagonal grid



# Sure, bins, but where are my cells?

A bin does not constitute a cell. But we have the chance to find and save the locations and extents of cells and find the bins inside. They can be aggregated and analyzed as a nucleus or cell if available.



# Coordinates, bins, polygons

The **purple** and **pink** pixels represent the histology, the illustration shows an example of a nucleus. In this case, every pixel covers a  $2\mu\text{m}$  bin. Every bin and pixel has an  $(x, y)$  coordinate so we know where it is located in space and we know who its neighbors are.

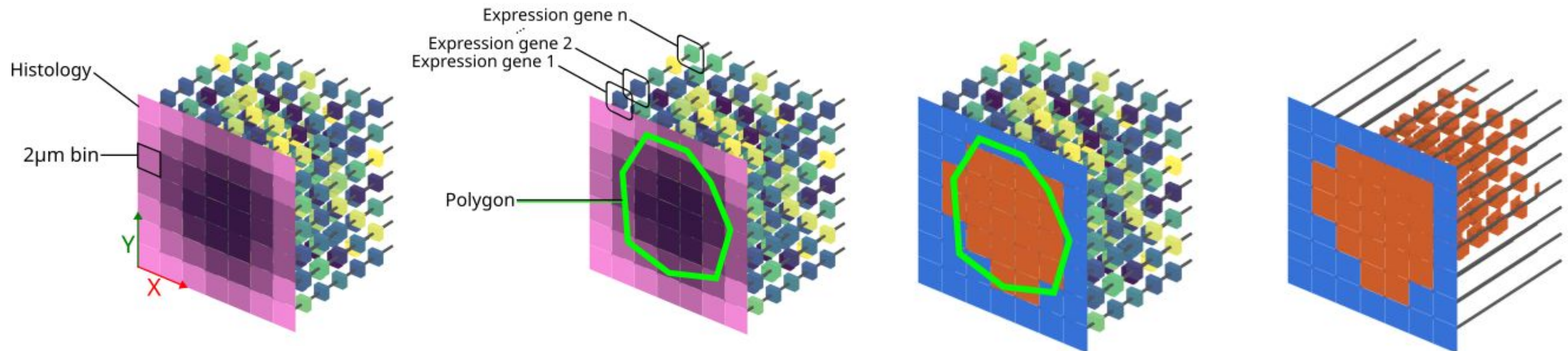
Every cube colored in a **viridis** scale represents the  $\log(\text{norm})$  expression of a gene. There are 18k+ gene expression values per  $2\mu\text{m}$  bin.

A **green** polygon shows what an annotation delineates as a nucleus. Without this polygon we would not know the extent of the nucleus. It can be selected by a human expert observer or automatically via segmentation algorithms. They are stored either as polygons (geometry) or labels (pixel-like).

The polygon can also be converted into a label, illustrated here as **orange** and **blue**. Both Polygons and Labels can be stored. Both humans and algorithms can draw polygons or paint pixels.

The expression accumulated within a label is what goes further into data analysis.

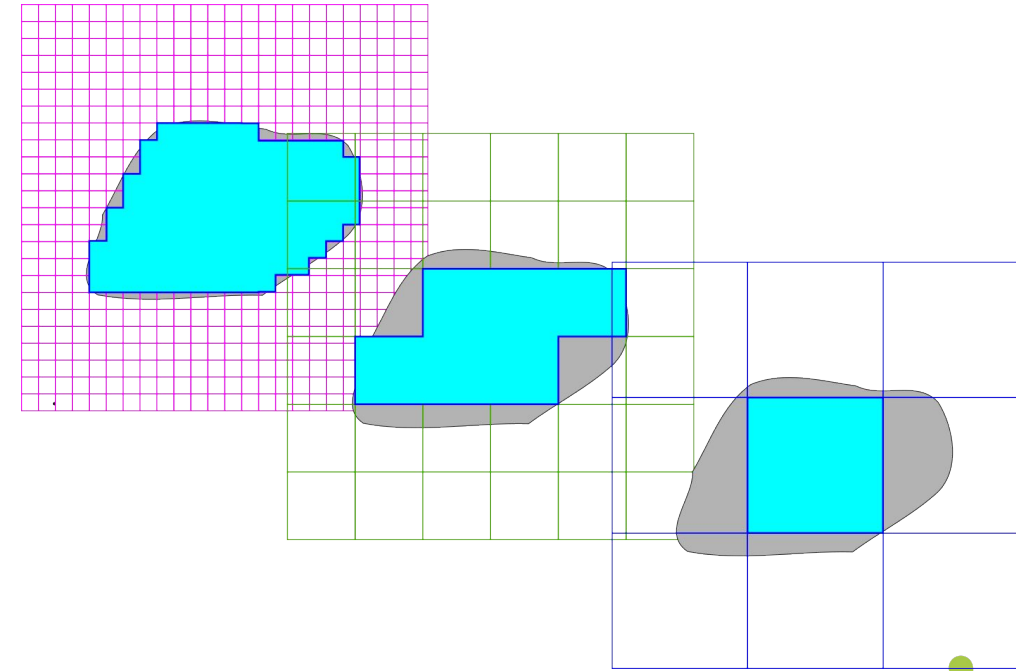
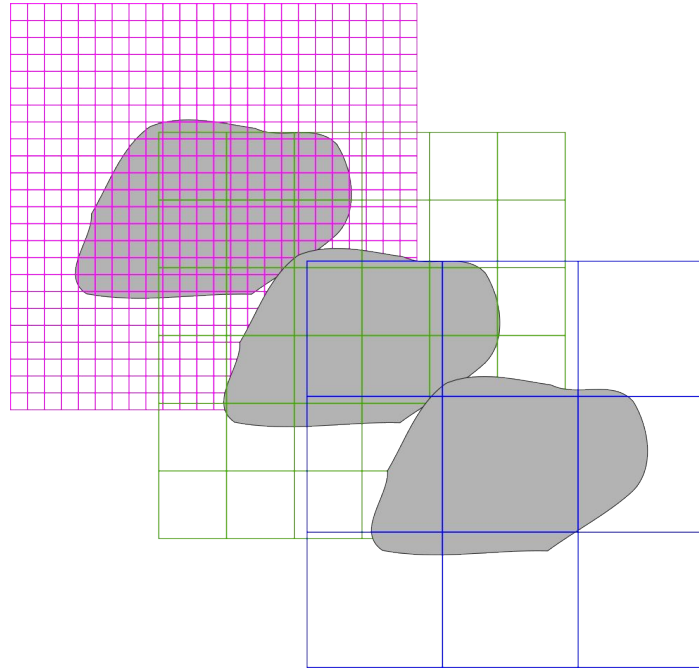
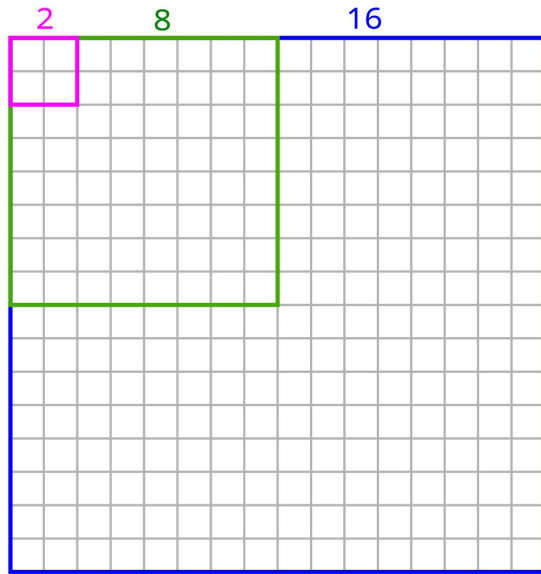
Individual expression vectors can also go to analysis without being aggregated. However, since references are more on a single-cell level, individual vectors are more difficult to pair with it. When no reference is used then it is also called reference-free and segmentation-free analysis.



# Sequence based

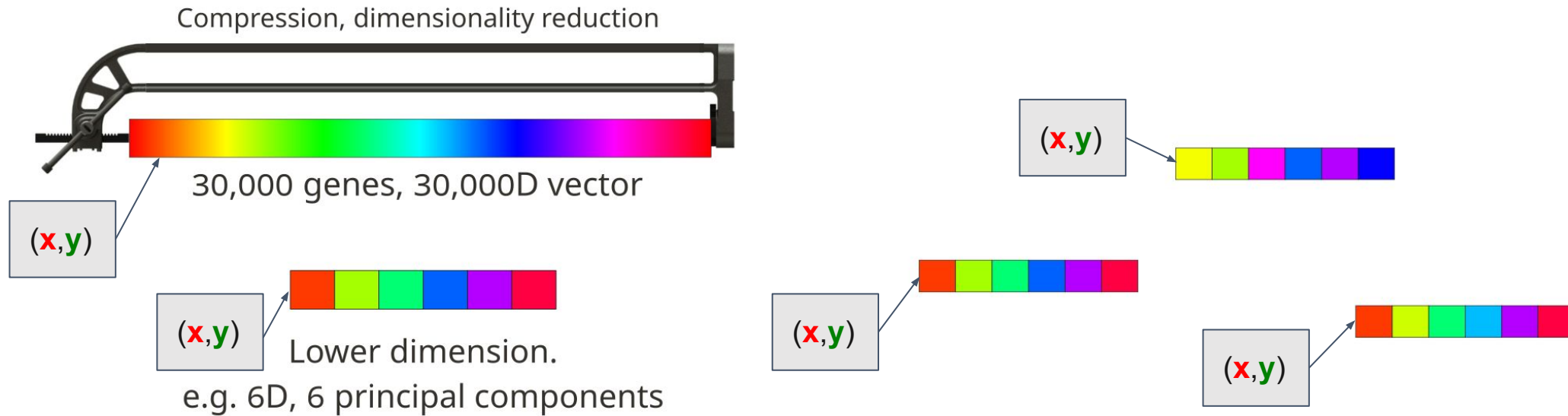
Sometimes our samples have cells and or nuclei already segmented, meaning polygons and/or labels exist. We can aggregate the bins inside. Tables with the aggregated bins are already computed.

If we don't have cell/nucleus segmentation we can also take individual bins. Visium HD, specially in later versions, comes with cell/nucleus segmentation. But it also comes with 2um bins, 8um bins and 16um bins.



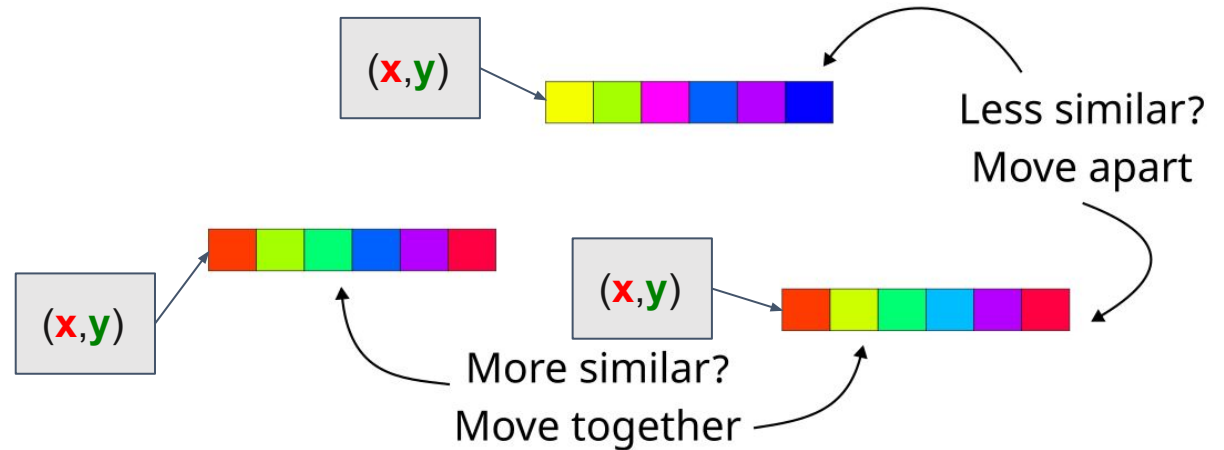
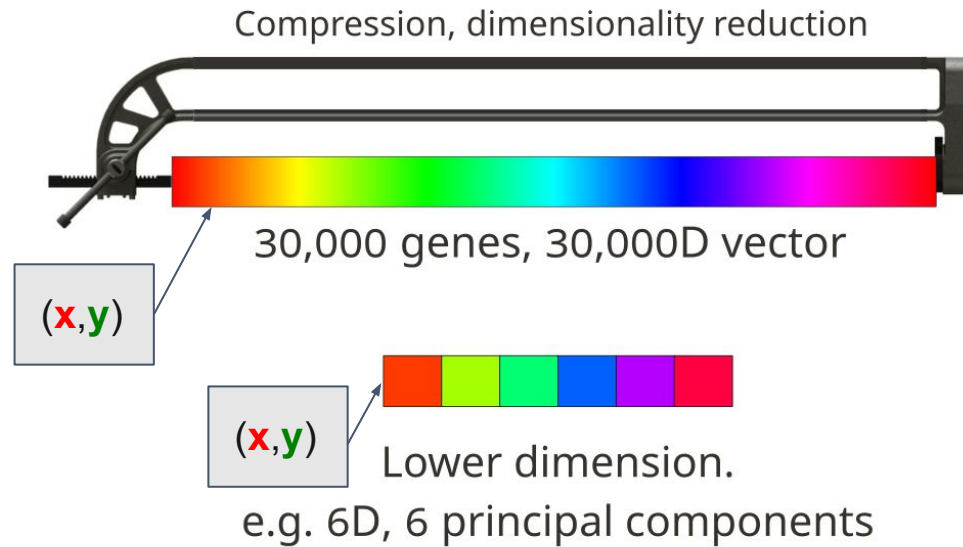
# Cells or bins, all is vectors

Most atlases are made for cell-level data, so it is always better to have cell segmentation. However, segmentation accuracy matters in the final results. If no segmentation is available, clustering still works, but the smaller the bin, the noisier the clustering will be.



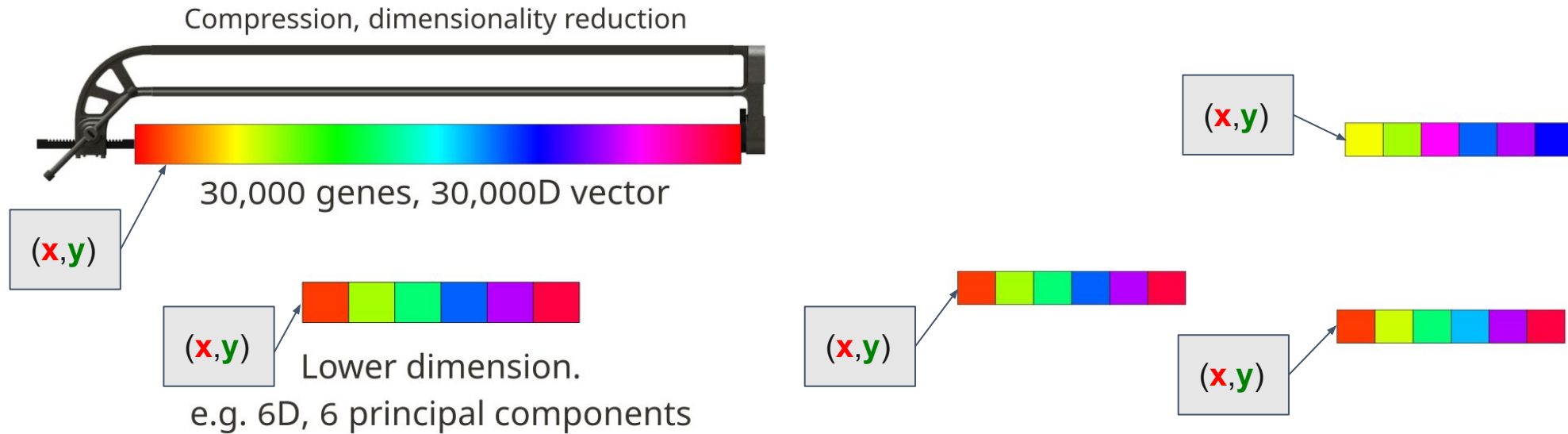
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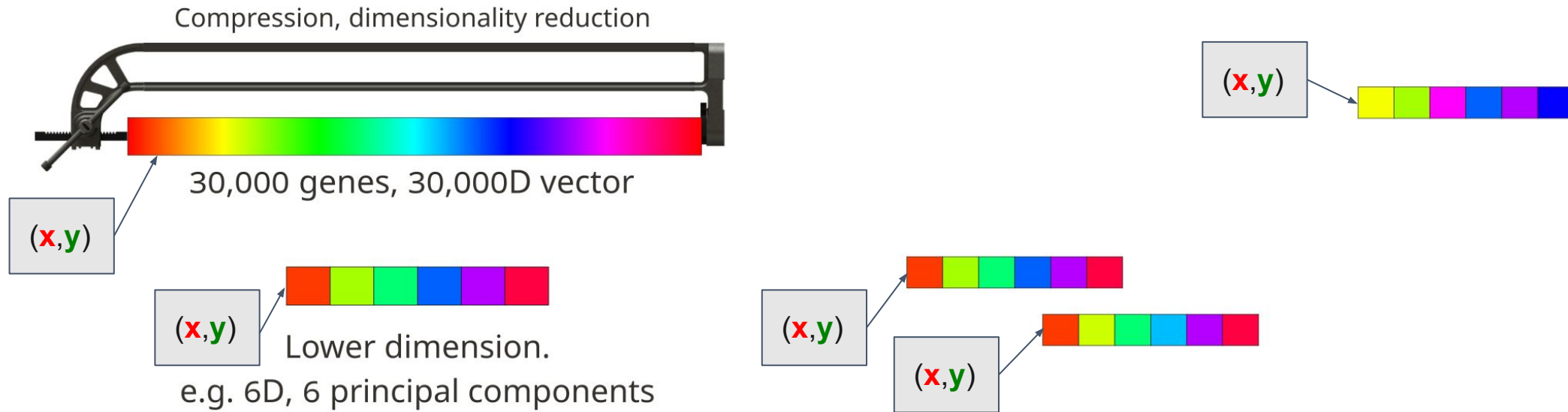
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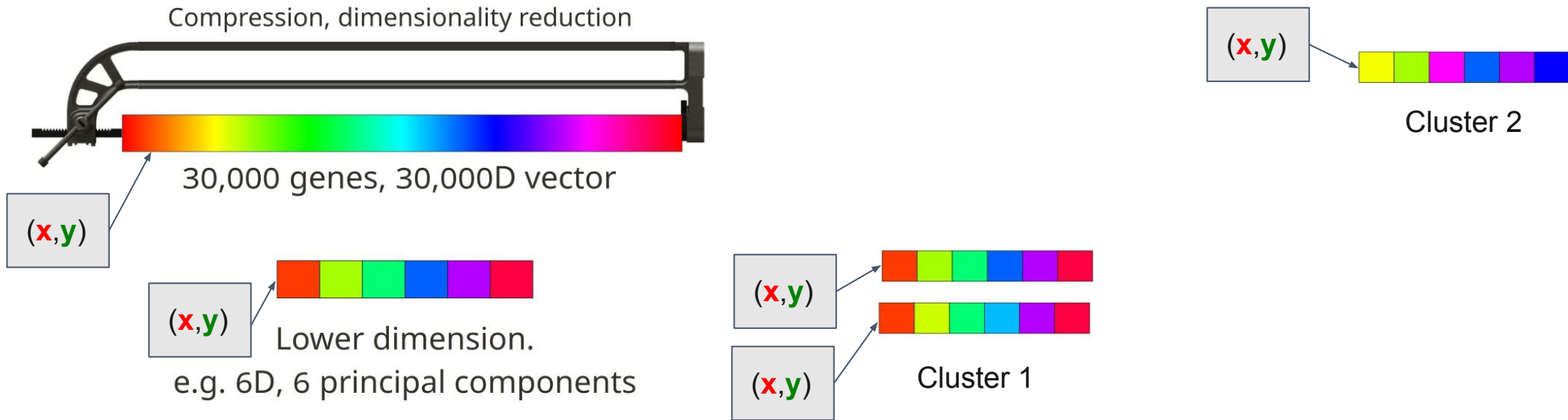
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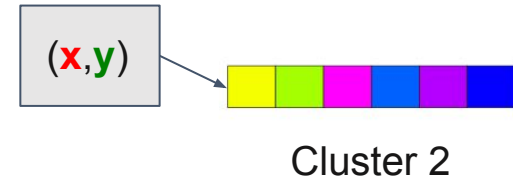
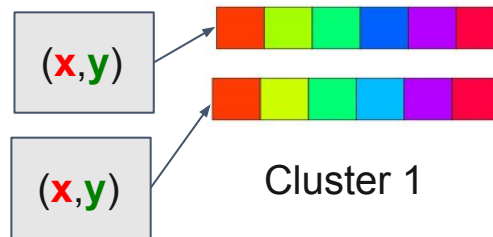
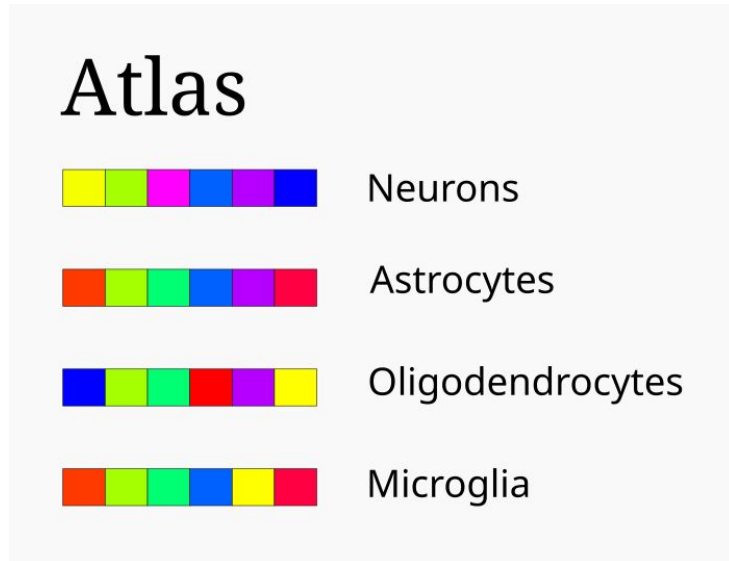
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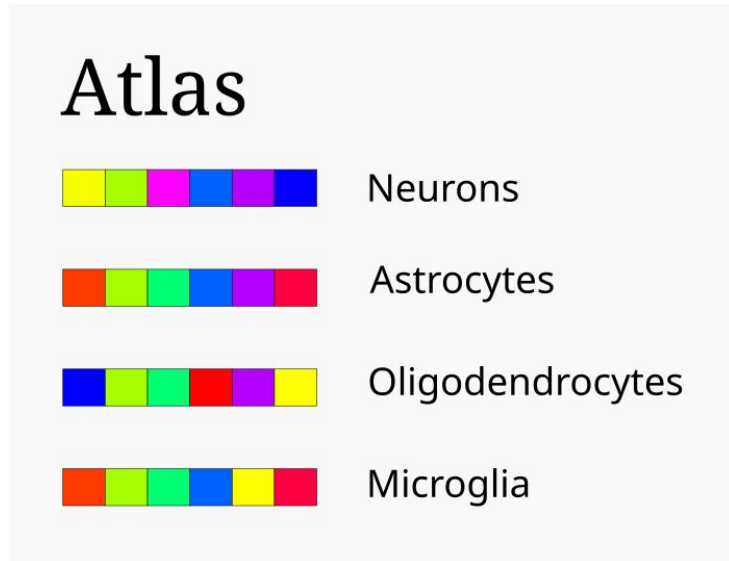
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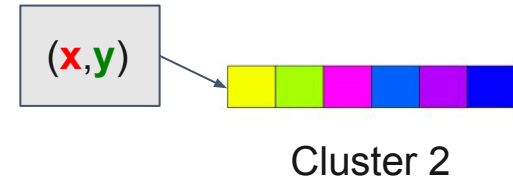
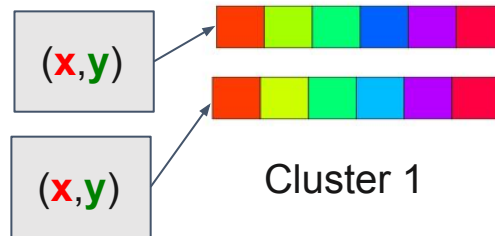


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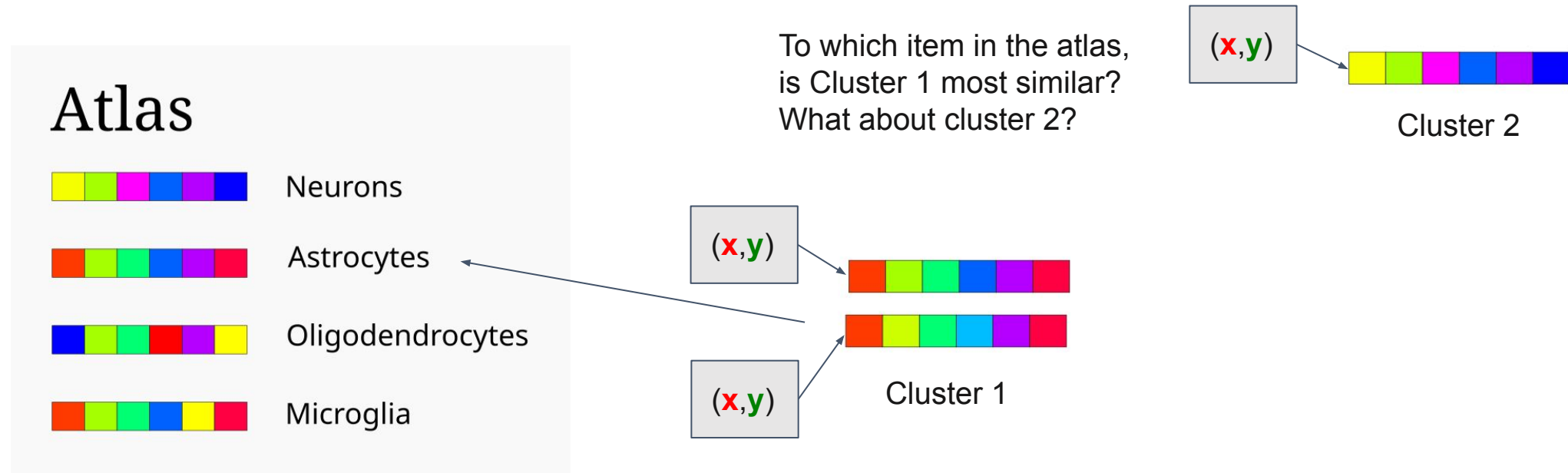


To which item in the atlas,  
is Cluster 1 most similar?  
What about cluster 2?



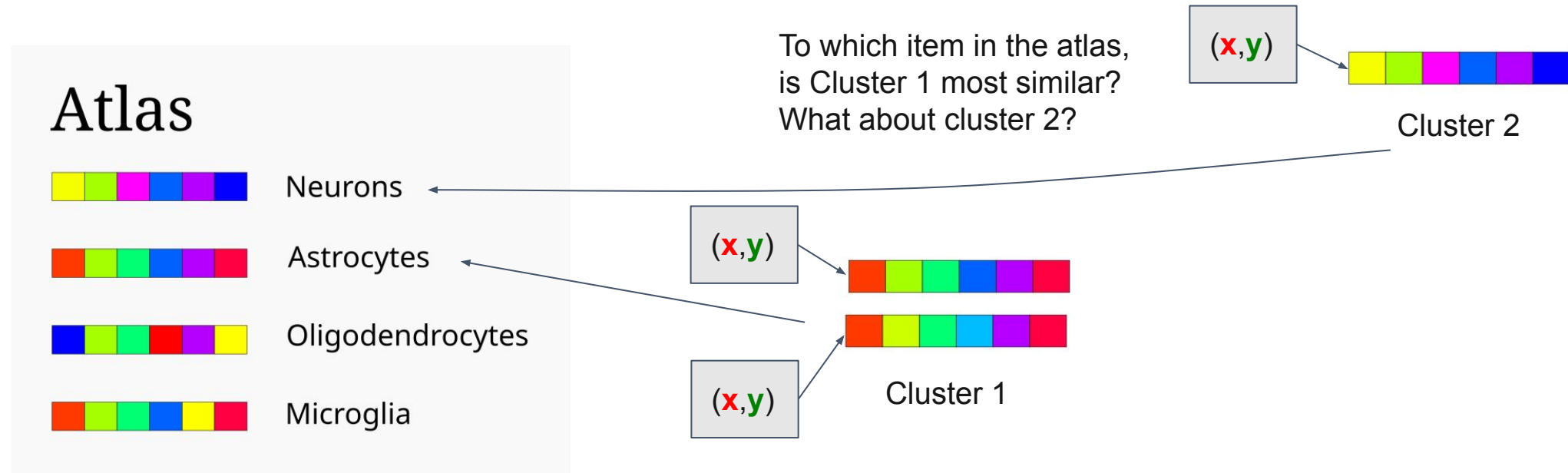
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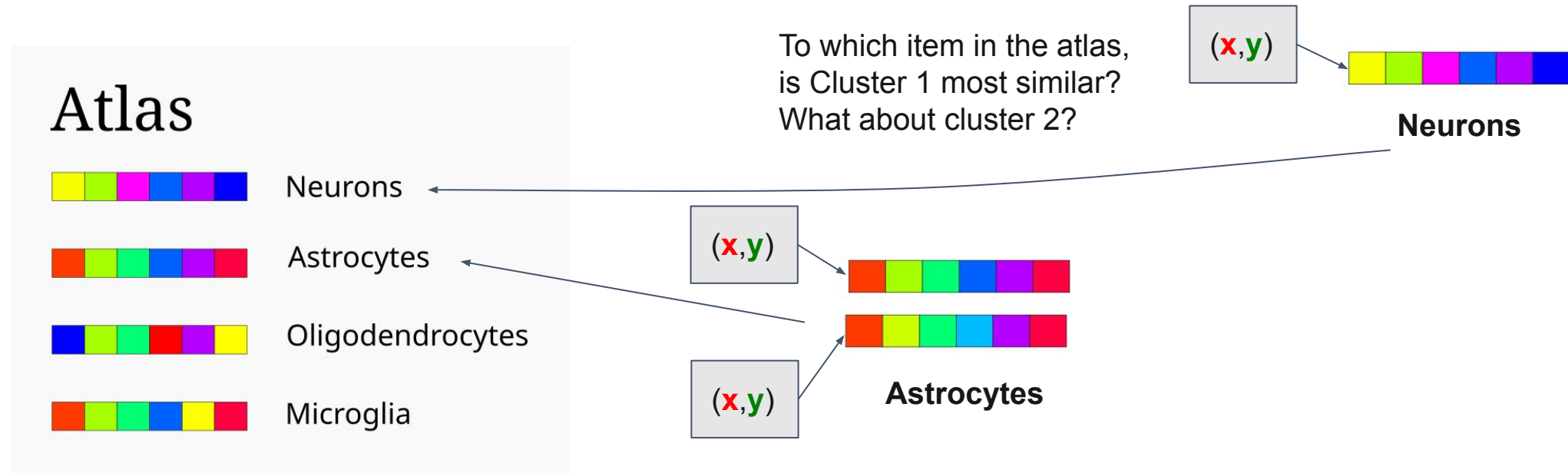
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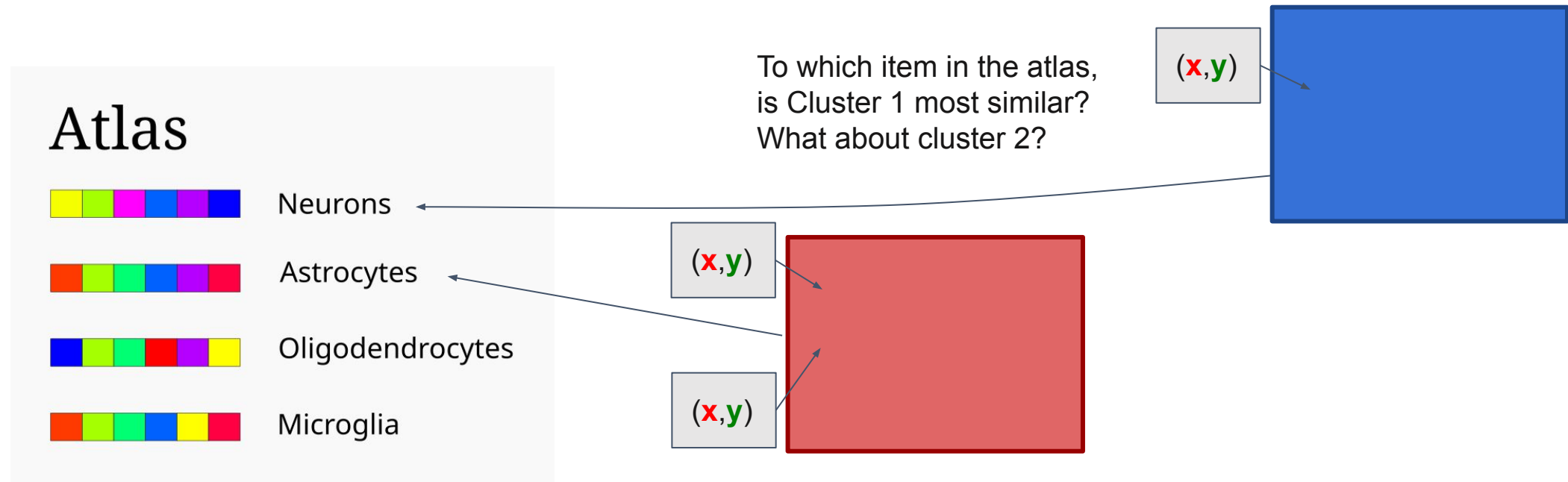
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# Map it back!

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Since all bins have a ( $x, y$ ) coordinate (location in space), we can map their clusters back to space and we can see where cells are located and see them in context, observe the structures they form and the most common relationships or groups of cell types.

